UNIVERSITY — ACADEMIC COURSE SYLLABUS

UE24CS2404

Unstructured Data & NoSQL Technologies

Program:	B.Tech Computer Science
Department:	Computer Science & Engineering
Semester:	Semester 4
Credits:	4 | 60 Hours
Difficulty:	Easy
Assessment:	CIE 1(20) + CIE 2(20) + SEE(60)

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1. Course Overview

Field	Details
Course Code	UE24CS2404
Course Title	Unstructured Data & NoSQL Technologies
Program	B.Tech Computer Science
Department	Computer Science & Engineering
Semester	Semester 4
Credits	4
Total Hours	60
Difficulty	Easy

2. Course Objectives

1. Understand foundational concepts of Unstructured Data & NoSQL Technologies.
2. Apply techniques of Unstructured Data & NoSQL Technologies to real-world problems.
3. Evaluate and design solutions using Unstructured Data & NoSQL Technologies.

3. Course Outcomes (COs)

CO ID	Description	Bloom's Level
CO1	Recall and explain key concepts of Unstructured Data & NoSQL Technologies.	Remember
CO2	Apply core methods of Unstructured Data & NoSQL Technologies in problem-solving.	Apply
CO3	Analyze and evaluate Unstructured Data & NoSQL Technologies scenarios.	Analyze
CO4	Design solutions integrating Unstructured Data & NoSQL Technologies principles.	Create

4. CO-PO Mapping

Mapping Scale: 3 = High | 2 = Medium | 1 = Low | — = Not Mapped

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	—	—	—	—	—	—	—	—	1	2	1
CO2	2	3	2	1	—	—	—	—	—	—	—	1	3	2
CO3	2	2	3	2	1	—	—	—	—	—	1	2	2	3
CO4	1	2	3	3	2	1	—	—	1	1	2	2	2	3

5. Syllabus – Unit-wise Content

Unit 1: Introduction to Unstructured Data & NoSQL Technologies (12 Hours)

#	Topic
1	Basic Concepts
2	History & Evolution
3	Core Principles
4	Terminology

Unit 2: Core Techniques (15 Hours)

#	Topic
1	Fundamental Methods
2	Implementation Strategies
3	Algorithms
4	Tools

Unit 3: Advanced Topics (15 Hours)

#	Topic
1	Advanced Concepts
2	Case Studies
3	Performance Analysis
4	Optimization

Unit 4: Applications & Projects (9 Hours)

#	Topic
1	Real-World Applications
2	Mini Projects
3	Problem Solving

Unit 5: Emerging Trends (9 Hours)

#	Topic
1	Current Research
2	Industry Practices
3	Future Directions

6. Lesson Plan

Week	Lecture	Topics Covered	COs	Method	Hrs
1	L1-L2	Introduction to Unstructured Data & NoSQL Technologies	CO1	Lecture	2
2	L3-L4	Core Concepts & Fundamentals	CO1, CO2	Lecture + Demo	2
3	L5-L6	Implementation Techniques	CO2	Lab Session	3
4	L7-L8	Problem Solving Workshop	CO2, CO3	Workshop	2
5	L9	CIE 1 Revision	CO1, CO2	Revision	1
6	L10-L11	Advanced Topics Part I	CO3	Lecture	2
7	L12-L13	Advanced Topics Part II	CO3, CO4	Flipped Classroom	2
8	L14-L15	Design & Build Lab	CO4	Lab Session	3
9	L16	CIE 2 Revision	CO3, CO4	Revision	1
10	L17-L18	Applications & Case Studies	CO4	Case Study	2

7. Textbooks & References

7.1 Textbooks

#	Title	Author(s)	Publisher	Year
1	Fundamentals of Unstructured Data & NoSQL Technologies	John D. Smith	Pearson Education	2022
2	Unstructured Data & NoSQL Technologies: A Practical Approach	Robert A. Kumar	McGraw-Hill	2021

7.2 Reference Books

#	Title	Author(s)	Publisher	Year
1	Advanced Unstructured Data & NoSQL Technologies	Jane M. Lee	Wiley	2020
2	Unstructured Data & NoSQL Technologies in Practice	S. K. Gupta	Oxford Press	2019

8. Assessment Scheme

Component	Max Marks	Description
CIE 1	20	1st Continuous Internal Evaluation
CIE 2	20	2nd Continuous Internal Evaluation
SEE	60	Semester End Examination

9. Grading Scheme & Attainment Targets

9.1 Grading Scheme

Absolute Grading

9.2 Attainment Targets

80% of students scoring > 60%

9.3 CO-PO Attainment Summary

CO	Bloom's Level	Assessment Method	Target Attainment
CO1	Remember	CIE + SEE	"e 60%
CO2	Apply	CIE + SEE	"e 60%
CO3	Analyze	CIE + SEE	"e 60%
CO4	Create	CIE + SEE	"e 60%

— End of Course Syllabus Document —

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